Worksheet 3.3 — Solving Rational Equations & Inequalities

Grade 12 Advanced Functions (MHF4U) · Unit 3: Rational Functions

The Big Ideas

Clear fractions with the LCD and check restrictions; for inequalities, get 0 on one side, find the critical values (zeros of the numerator and of the denominator) and use a **sign table**. In every step write the **whole equation or inequality**, even when you are only working on one term.

- Equation: state the restrictions (denominator $\neq 0$), multiply *every* term by the LCD, solve, then reject extraneous roots.
- Inequality: never multiply by an expression with x. Get 0 on one side, write one fraction, find the critical values, and build a sign table (top, bottom, whole fraction).
- Denominator zeros are always excluded (undefined); numerator zeros are included only for $\geq$ or $\leq$.

Core Concepts

Solving an equation

State the restrictions (every denominator $\neq 0$), find the LCD, and multiply *every* term on both sides by it. Solve what remains, then **check**: any answer that makes an original denominator zero is an *extraneous root* and must be rejected.

Solving an inequality

Never multiply by an expression containing x (its sign is unknown, so you cannot tell whether to flip the inequality). Instead get 0 on one side, combine into one fraction, and factor the top and bottom.

Critical values and the sign table

Zeros of the numerator make the fraction 0 (included only for $\geq$ or $\leq$). Zeros of the denominator make it undefined and are *never* included. Build a sign table with a row for each factor on top and bottom; the fraction's sign is the product of the signs.

Writing solutions

In every step write the **whole equation or inequality**, even when you are only working on one term or one denominator. For example: "multiply every term of $\dfrac{2}{x} + 1 = 3$ by x."

Worked Examples

Example 1 — Simple equation

Solve $\dfrac{1}{x} = 2$.

Solution. **Step 1:** The denominator of $\dfrac{1}{x} = 2$ is x, so $x \neq 0$.

Step 2: Multiply both sides of $\dfrac{1}{x} = 2$ by x: $1 = 2x$, so $x = \frac{1}{2}$.

Step 3: Check: $x = \frac{1}{2} \neq 0$ and $\dfrac{1}{1/2} = 2$ ✓.

Answer: $x = \frac{1}{2}$.

Example 2 — Cross-multiply

Solve $\dfrac{x+3}{x-1} = 4$.

Solution. **Step 1:** The denominator of $\dfrac{x+3}{x-1} = 4$ is $x-1$, so $x \neq 1$.

Step 2: Multiply both sides of $\dfrac{x+3}{x-1} = 4$ by $x-1$: $x+3 = 4(x-1)$.

Step 3: Expand $x+3 = 4(x-1)$: $x+3 = 4x-4$, so $7 = 3x$ and $x = \frac{7}{3}$.

Step 4: Check: $x = \frac{7}{3} \neq 1$, and $\dfrac{\frac{7}{3}+3}{\frac{7}{3}-1} = \dfrac{16/3}{4/3} = 4$ ✓.

Answer: $x = \frac{7}{3}$.

Example 3 — Clear the LCD

Solve $\dfrac{9}{x} + 1 = 4$.

Solution. **Step 1:** The denominator of $\dfrac{9}{x} + 1 = 4$ is x, so $x \neq 0$.

Step 2: Multiply *every* term of $\dfrac{9}{x} + 1 = 4$ by x: $9 + x = 4x$.

Step 3: From $9 + x = 4x$: $9 = 3x$, so $x = 3$. Check: $\dfrac{9}{3} + 1 = 4$ ✓.

Answer: $x = 3$.

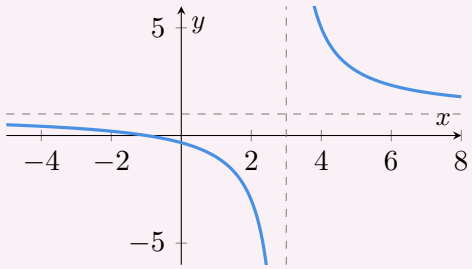
Example 4 — Inequality

Solve $\dfrac{x+1}{x-3} > 0$.

Solution. **Step 1:** Critical values of $\dfrac{x+1}{x-3}$: numerator $x+1=0 \Rightarrow x=-1$; denominator $x-3=0 \Rightarrow x=3$ (excluded).
Step 2: Sign table (the sign of the fraction is the sign of the top times the sign of the bottom):

Interval	$x<-1$	$x=-1$	$-1<x<3$	$x=3$	$x>3$
Test point	-2	–	0	–	4
top $x+1$	$-$	0	$+$	$+$	$+$
bottom $x-3$	$-$	$-$	$-$	0	$+$
$\dfrac{x+1}{x-3}$	$+$	0	$-$	und.	$+$
$\dfrac{x+1}{x-3}>0$?	✓	×	×	×	✓

Step 3: Keep the ✓ columns. At $x=-1$ the fraction is 0, not >0; at $x=3$ it is undefined.
Answer: $x<-1$ or $x>3$, i.e. $(-\infty,-1)\cup(3,\infty)$. The curve is above the axis there:



Example 5 — Reciprocal inequality

Solve $\dfrac{1}{x-5} < 0$.

Solution. **Step 1:** The numerator of $\dfrac{1}{x-5}$ is the positive constant 1, so only the denominator matters. Critical value: $x-5=0 \Rightarrow x=5$ (excluded).

Interval	$x<5$	$x=5$	$x>5$
Test point	4	–	6
top 1	$+$	$+$	$+$
bottom $x-5$	$-$	0	$+$
$\dfrac{1}{x-5}$	$-$	und.	$+$
$\dfrac{1}{x-5}<0$?	✓	×	×

Step 2: We want $\dfrac{1}{x-5} < 0$, so the bottom $x-5$ must be negative.
Answer: $x<5$, i.e. $(-\infty,5)$.

Example 6 — Cross-multiply

Solve $\dfrac{2x-1}{x+2}=3$.

Solution. **Step 1:** The denominator of $\dfrac{2x-1}{x+2}=3$ is $x+2$, so $x\neq -2$.

Step 2: Multiply both sides of $\dfrac{2x-1}{x+2}=3$ by $x+2$: $2x-1=3(x+2)=3x+6$.

Step 3: Solve: $-7=x$. Check: $x=-7\neq -2$, and $\dfrac{2(-7)-1}{-7+2}=\dfrac{-15}{-5}=3$ ✓.

Answer: $x=-7$.

Example 7 — Extraneous root

Solve $\dfrac{x^2}{x-3}=\dfrac{9}{x-3}$.

Solution. **Step 1:** The denominators of $\dfrac{x^2}{x-3}=\dfrac{9}{x-3}$ are $x-3$, so $x\neq 3$.

Step 2: Multiply both sides by $x-3$: $x^2=9$, so $x=3$ or $x=-3$.

Step 3: Check: $x=3$ breaks the restriction $x\neq 3$, so it is *extraneous*. For $x=-3$: $\dfrac{9}{-6}=-\frac{3}{2}$ and $\dfrac{9}{-6}=-\frac{3}{2}$ ✓.

Answer: $x=-3$ (reject $x=3$).

Example 8 — Inequality

Solve $\dfrac{x-4}{x}>0$.

Solution. **Step 1:** Critical values of $\dfrac{x-4}{x}$: numerator $x-4=0\Rightarrow x=4$; denominator $x=0$ (excluded).

Interval	$x<0$	$x=0$	$0<x<4$	$x=4$	$x>4$
Test point	-1	–	1	–	5
top $x-4$	$-$	$-$	$-$	0	$+$
bottom x	$-$	0	$+$	$+$	$+$
$\dfrac{x-4}{x}$	$+$	und.	$-$	0	$+$
$\dfrac{x-4}{x}>0$?	✓	×	×	×	✓

Step 2: Keep the ✓ columns.

Answer: $x<0$ or $x>4$, i.e. $(-\infty,0)\cup(4,\infty)$.

Example 9 — Get 0 on one side

Solve $\dfrac{x+3}{x-1} \le 2$.

Solution. **Step 1:** Do not multiply $\dfrac{x+3}{x-1} \le 2$ by $x-1$ (its sign is unknown). Subtract 2: $\dfrac{x+3}{x-1} - 2 \le 0$.

Step 2: One fraction: $\dfrac{x+3-2(x-1)}{x-1} \le 0$, so $\dfrac{5-x}{x-1} \le 0$.

Step 3: Critical values: numerator $5-x=0 \Rightarrow x=5$ (included, because of $\le$); denominator $x-1=0 \Rightarrow x=1$ (excluded).

Interval	$x<1$	$x=1$	$1<x<5$	$x=5$	$x>5$
Test point	0	–	2	–	6
top $5-x$	+	+	+	0	−
bottom $x-1$	−	0	+	+	+
$\dfrac{5-x}{x-1}$	−	und.	+	0	−
$\dfrac{5-x}{x-1} \le 0$?	✓	×	×	✓	✓

Step 4: Keep the − columns and the zero of the numerator.
Answer: $x<1$ or $x \ge 5$, i.e. $(-\infty, 1) \cup [5, \infty)$.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Solve $\frac{7}{x} = 2$.

Question 2

Solve $\frac{x+4}{x-1} = 5$.

Question 3

Solve $\frac{10}{x} - 3 = 2$.

Question 4

Solve $\frac{x}{x+6} > 0$.

Question 5

Solve $\frac{-1}{x+2} > 0$.

Question 6

Solve $\frac{5}{x} = 10$.

Question 7

Solve $\frac{x+2}{x-1} = 3$.

Question 8

Solve $\frac{6}{x} + 2 = 5$.

Question 9

Solve $\frac{x}{x-3} < 0$.

Question 10

Solve $\dfrac{1}{x-4} > 0$.

Question 11

Solve $\dfrac{2}{x} = \dfrac{1}{x-1}$.

Question 12

Solve $\dfrac{x-2}{x+1} \geq 0$.

Question 13 — Challenge

Solve $\dfrac{x+1}{x-2} \leq 0$ using critical values.

Question Recap & Answer Key

Question Recap

1. Solve $\frac{7}{x} = 2$.
2. Solve $\frac{x+4}{x-1} = 5$.
3. Solve $\frac{10}{x} - 3 = 2$.
4. Solve $\frac{x}{x+6} > 0$.
5. Solve $\frac{-1}{x+2} > 0$.
6. Solve $\frac{5}{x} = 10$.
7. Solve $\frac{x+2}{x-1} = 3$.
8. Solve $\frac{6}{x} + 2 = 5$.
9. Solve $\frac{x}{x-3} < 0$.
10. Solve $\frac{1}{x-4} > 0$.
11. Solve $\frac{2}{x} = \frac{1}{x-1}$.
12. Solve $\frac{x-2}{x+1} \geq 0$.
13. Solve $\frac{x+1}{x-2} \leq 0$ using critical values.

✓ Answer Key

1. For $\frac{7}{x} = 2$: $x \neq 0$; multiply both sides by x: $7 = 2x$, so $x = \frac{7}{2}$ (check: $\frac{7}{7/2} = 2$)
2. For $\frac{x+4}{x-1} = 5$: $x \neq 1$; multiply both sides by $x-1$: $x+4 = 5(x-1) = 5x-5$, so $9 = 4x$ and $x = \frac{9}{4}$ (check: $\frac{25/4}{5/4} = 5$)
3. For $\frac{10}{x} - 3 = 2$: $x \neq 0$; multiply every term by x: $10 - 3x = 2x$, so $10 = 5x$ and $x = 2$ (check: $\frac{10}{2} - 3 = 2$)
4. For $\frac{x}{x+6} > 0$: critical values $-6, 0$ (bottom zeros excluded); signs $+, -, +$; $x < -6$ or $x > 0$, i.e. $(-\infty, -6) \cup (0, \infty)$
5. For $\frac{-1}{x+2} > 0$: critical values -2 (bottom zeros excluded); signs $+, -$; $x < -2$, i.e. $(-\infty, -2)$
6. For $\frac{5}{x} = 10$: $x \neq 0$; multiply both sides by x: $5 = 10x$, so $x = \frac{1}{2}$ (check: $\frac{5}{1/2} = 10$)
7. For $\frac{x+2}{x-1} = 3$: $x \neq 1$; multiply both sides by $x-1$: $x+2 = 3(x-1) = 3x-3$, so $5 = 2x$ and $x = \frac{5}{2}$ (check: $\frac{9/2}{3/2} = 3$)
8. For $\frac{6}{x} + 2 = 5$: $x \neq 0$; multiply every term by x: $6 + 2x = 5x$, so $6 = 3x$ and $x = 2$ (check: $\frac{6}{2} + 2 = 5$)
9. For $\frac{x}{x-3} < 0$: critical values $0, 3$ (bottom zeros excluded); signs $+, -, +$; $0 < x < 3$, i.e. $(0, 3)$
10. For $\frac{1}{x-4} > 0$: critical values 4 (bottom zeros excluded); signs $-, +$; $x > 4$, i.e. $(4, \infty)$
11. For $\frac{2}{x} = \frac{1}{x-1}$: $x \neq 0, 1$; multiply every term by the LCD $x(x-1)$: $2(x-1) = x$, so $2x - 2 = x$ and $x = 2$ (check: $\frac{2}{2} = 1$ and $\frac{1}{2-1} = 1$)
12. For $\frac{x-2}{x+1} \geq 0$: critical values $-1, 2$ (bottom zeros excluded); signs $+, -, +$; $x < -1$ or $x \geq 2$, i.e. $(-\infty, -1) \cup [2, \infty)$
13. For $\frac{x+1}{x-2} \leq 0$: critical values $-1, 2$ (bottom zeros excluded); signs $+, -, +$; $-1 \leq x < 2$, i.e. $[-1, 2)$